Gradient Descent (GD) – Stochastic Gradient Descent SGD) and Mini-Batch Gradient Descent

- Stochastic Gradient Descent (SGD) is a variant of GD which computes the gradient of the Error function with respect to a single training example
- In other words, it updates the weights using one training sample at a time
- Remember: GD updated the weight vector after passing through ALL training samples.

$$W^{(updated)} = W^{old} - \eta \delta_i X_i$$ Repeat for ALL misclassified Samples

Why using SGD

- SGD is usually more efficient than GD
- SGD converges faster than GD

Batch GD is a generalization of GD and SGD where weights are updated after passing through "Mini-Batch" of training samples with size n

If n=N (it is GD), and if n=1 (it is SGD)